CMU-AFRICA

MSEAI CAPSTONE PROJECT

Safety Data Collection and Processing System

For Rwanda Civil Aviation Authority

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December 2025

The Problem

East African Civil Aviation Authorities classify and analyse aviation safety reports manually, which is slow, inconsistent, and limits data-driven trend detection and ICAO Annex 19 compliance. Based on this analysis, we built an automated, accurate system that can reliably map free-text safety narratives to ICAO ADREP categories at scale

800+
Fatal accidents since 2020

30%
Analyst disagreement rate

48hr
Processing delay per report

ICAO Annex 19 Mandate

All member states MUST implement data-driven safety management. Manual processing cannot meet requirements.

Every misclassified report is a missed pattern. Missed patterns cost lives.

Research: NLP classification achieves 90%+ accuracy (New & Wallace, 2025)

Rwanda's Challenge

380
Reports/year

760
Hours wasted yearly

0
Automation

N/A
ICAO status

Current State

SafetyCore manual processing, inconsistent, delayed

Required State

Automated, real-time, ICAO compliant

RCAA needs reports processed in seconds, not days.

Our Solution: SDCPS Architecture

Data Ingestion
68,000 records

→

BERT Models
2 classifiers

→

LLM Validation
3 instances

→

Dashboard
Real-time

3,600	11	2.15s	95%
Validated	Categories	Processing	Auto-accept

Innovation: First multi-LLM consensus validation for aviation safety

Stack: PostgreSQL | FastAPI | Django | Docker | BERT | Gemini 2.5-Flash

Data Engineering

~ 68,000

Total Records Collected

ASRS (NASA)	ASN (Scraped)	PCI(SCraped)	RCAA
24,000	23,567	20,610	600

Database Design

PostgreSQL, 19 columns, optimized

Data Pipeline

ETL, cleaning, normalization

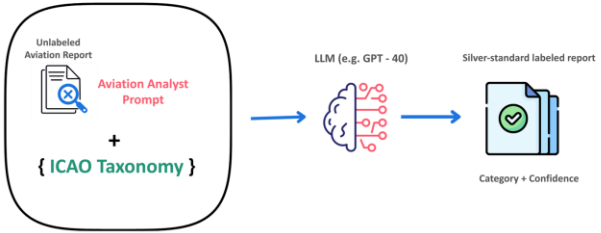
Web Scraping

Custom ASN extractors

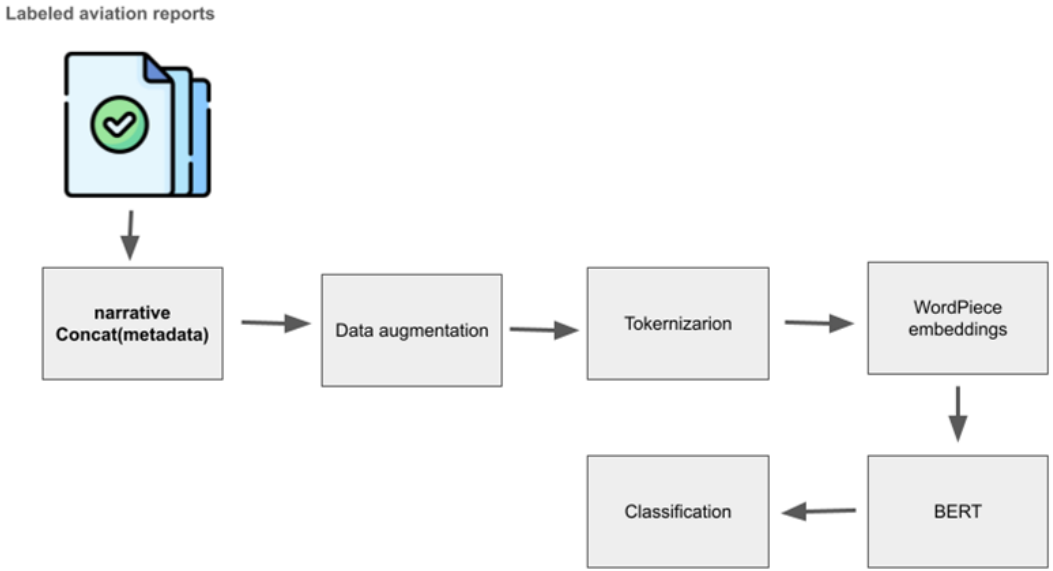
Impact: 100x more data than original RCAA dataset (600 → 76,190)

BERT - ASN

1

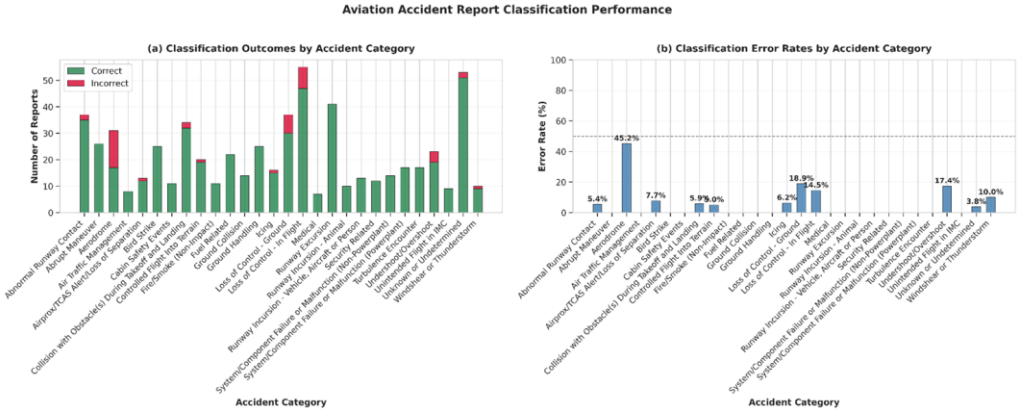

Unlabeled
Aviation Report
Aviation Analyst
Prompt
+
{ ICAO Taxonomy }
LLM (e.g. GPT - 40)
Silver-standard labeled report
Category + Confidence

2


Labeled aviation reports
narrative
Concat(metadata)
Data augmentation
Tokernizarion
WordPiece
embeddings
BERT
Classification

BERT - Cont.

1



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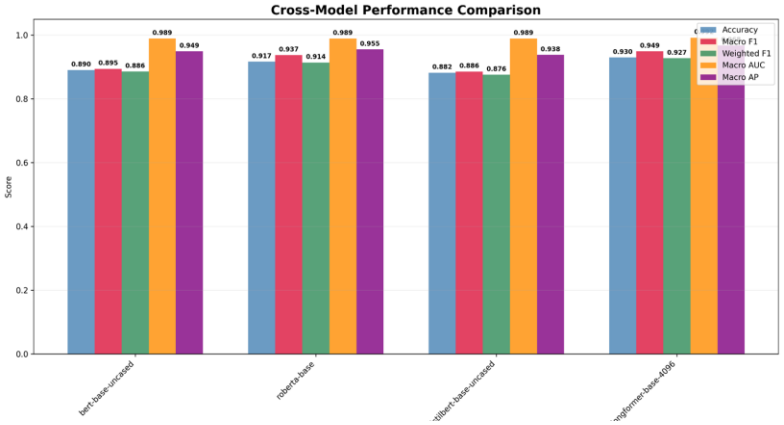


3

Table 1: Comparison of studies on aviation safety report classification

Study	Dataset	Categories	Model	Best Accuracy	Best Macro F1	Samples
Kuhn (2018)	NASA ASRS	Ad-hoc (12)	CNN	78.3%	0.74	5,626
Madeira et al. (2021)	Multiple	Severity (4)	LSTM	82.1%	0.79	1,240
New and Wallace (2025)	ASRS	Ad-hoc (30)	BERT-base	91.0%	0.91	425
This Study	**ASN**	**ICAO ADREP (28)**	**Longformer**	**92.96%**	**0.9493**	**3,068**

New, M. D., & Wallace, R. J. (2025). Classifying aviation safety reports: Using supervised natural language processing (NLP) in an applied context. *Safety*,

Zero Shot Learning with SafeAeroBERT Model

1. Semantic Embedding of Safety Narratives

- NASA's SafeAeroBERT model generates 512-dimension vectors for

2. Embedding ICAO ADREP Categories and Definitions

- Creates semantic anchors for ADREP categories using Hyper-Definitions

3. Computing Similarity

- Compare against 35 pre-embedded ICAO ADREP categories and Cosine similarity finds closest k=3 matches

4. Boosting & Safeguards

- Apply human expert rules to fix AI misinterpretations

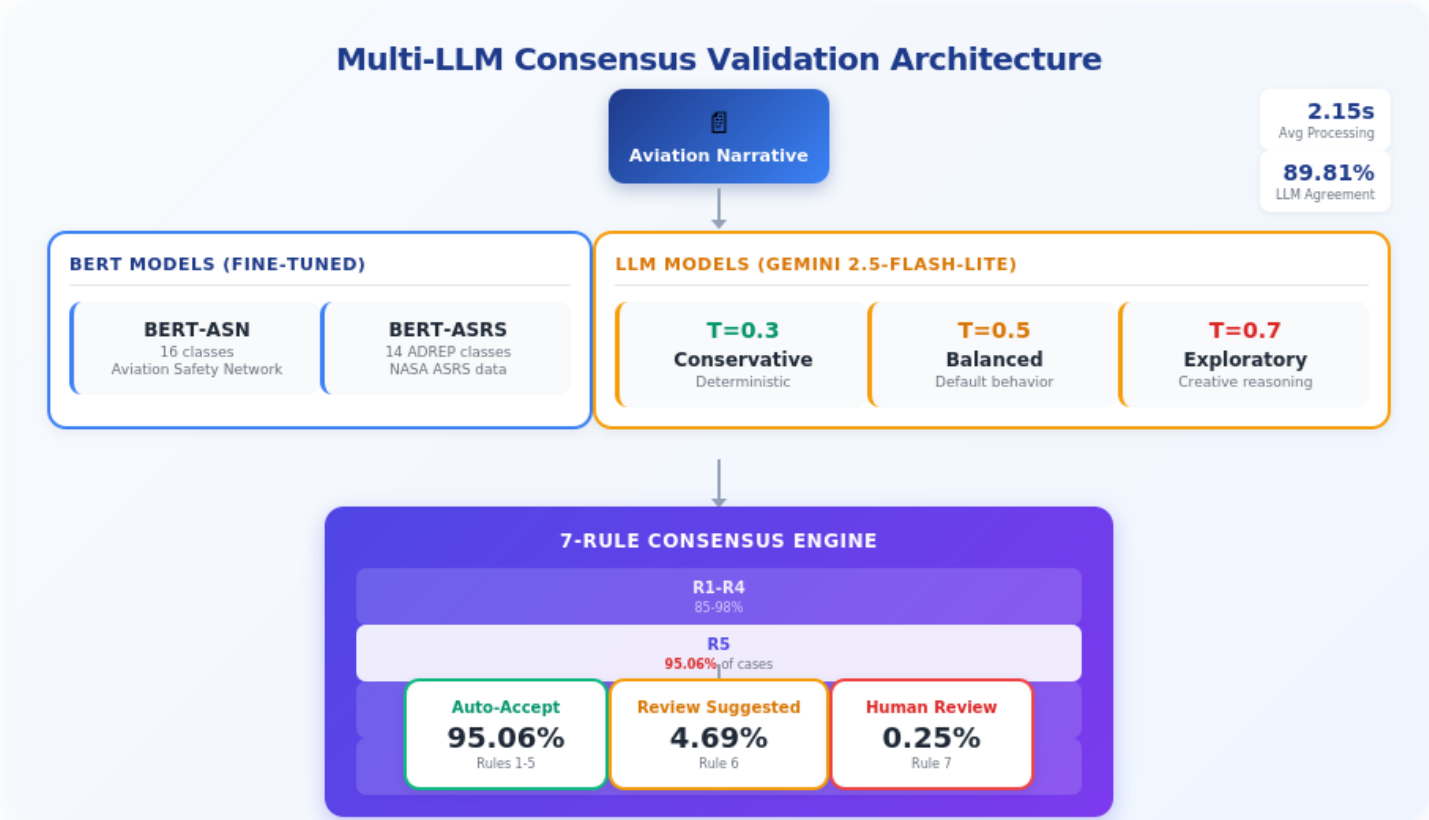
Distribution Transformation

Category	Base Bert	Hybrid (Bert + Boosting + Logic)
LOC-I	72% (1,408 reports)	<5%
AMAN	0.50%	<3%
WSTRW	<1%	4.3% (106 reports)
UIMC	<1%	19.5% (475 reports)

Top K Accuracy

ADREP Category	Top-1 Accuracy	Top-3 Accuracy	Top-5 Accuracy	Most Common Confusions
MAC (Mid-Air Collision)	87.5%	90.2%	96.3%	NMAC, AMAN
UIMC (Unintended IMC)	85.2%	91.8%	95.2%	WSTRW, SCF-NP
GCOL (Ground Collision)	83.1%	89.6%	92.7%	MAC, RAMP
RAMP (Ramp Event)	81.4%	88.9%	90.8%	GCOL, FUEL
WSTRW (Windshear)	79.4%	92.3%	94.1%	UIMC, TURB
SCF-NP (System Failure)	76.3%	91.7%	96.4%	LOC-I, USOS
LOC-I (Loss of Control)	72.6%	88.9%	93.5%	SCF-NP, AMAN
AMAN (Abrupt Maneuver)	68.9%	85.4%	92.1%	LOC-I, MAC

Multi-LLM Consensus Validation



Results: Exceeding All Targets

Metric	Target	Achieved	Status
Processing Speed	5s	**2.15s**	✓ 57% faster
Automation Rate	90%	**95.06%**	✓ Exceeded
Records Processed	380	**3,600**	✓ 9.5x more
LLM Agreement	80%	**89.81%**	✓ Exceeded

86,400×

Faster than manual (48hrs → 2s)

LIVE DEMONSTRATION

System Running Now

http://172.29.98.161:8888

3,600	11	LIVE
Records	Categories	Deployed

Dashboard shows real-time classification, category distribution, and confidence

Limitations

1. OTHER Category: 27%

Too many reports classified as OTHER reduces value.

Next: Manual review, prompt engineering

2. No Ground Truth

Agreement ≠ accuracy. We measure consensus.

Next: Expert-labeled test set (100+)

3. Rule 5 Dominance: 95%

LLMs override BERTs — may indicate training gaps.

Next: BERT retraining

Future

Validation

Expert labels, OTHER investigation

Integration

SafetyCore API, real-time ingestion

Enhancement

Predictive analytics, forecasting

Technical Improvements

BERT Retraining: Unified taxonomy

LLM Diversity: Claude/GPT-4

Upload: Real-time interface

IN SUMMARY

We Built a Production-Ready System

86,400×	95%	3,600	LIVE
Faster	Automation	Classified	Deployed

Nasiru Iliya | Ronnie Delyon | Baraka Manyara | John Bosco Thuo

Questions?

SAFE AIR

Dashboard | Evaluation

Evaluator: **RONNIE** Logout

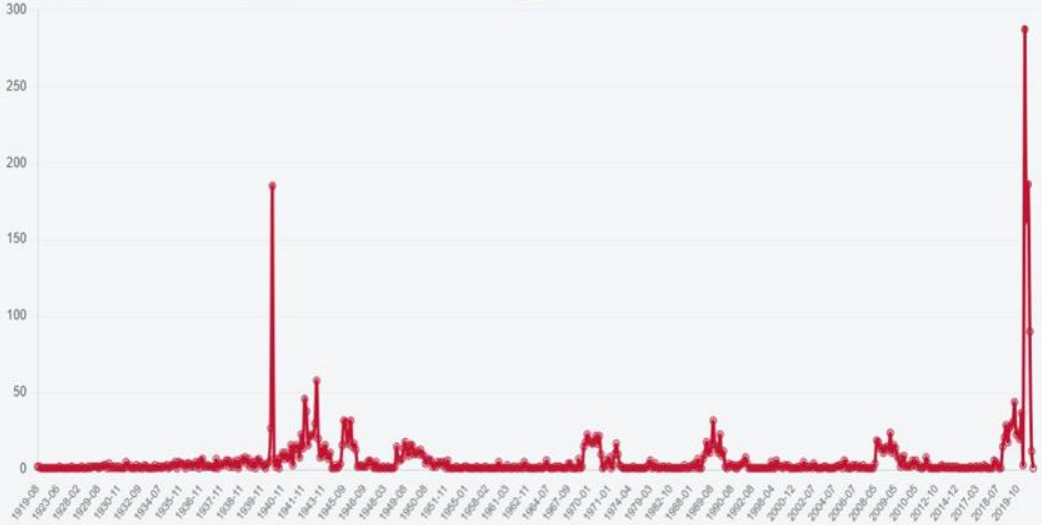
SAFETY ANALYTICS

- Dashboard
- Reports
- Evaluation

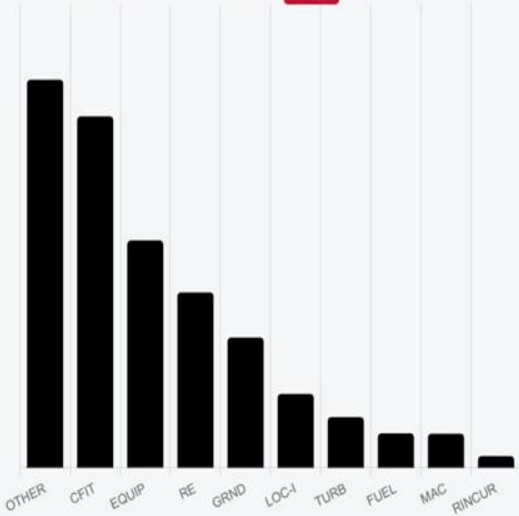
TOTAL INCIDENTS	FATAL INCIDENTS	TOTAL FATALITIES	GROUND INCIDENTS	SEVERITY SPLIT
46604	**0**	**0**	**0**	**0 Fatal / 46604 Non-Fatal**
↑ +5% this month	Critical Priority	Across all regions	Taxi/Ramp	

Incidents Over Time

All Classifications | All Phases



Incidents by Classification



Geographic Distribution

Incidents by Location

Loc

SAFE AIR

Dashboard | Evaluation

Evaluator: **RONNIE** Logout

SAFETY ANALYTICS

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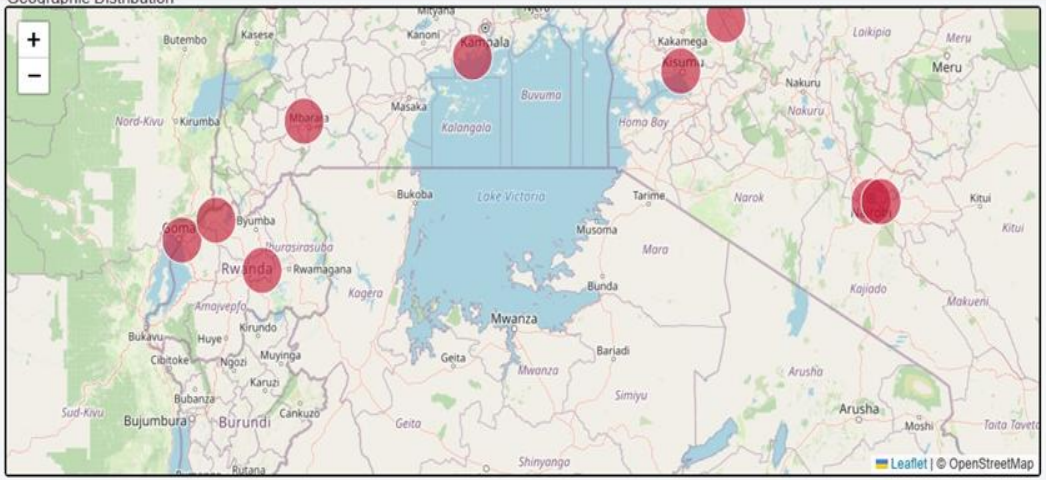
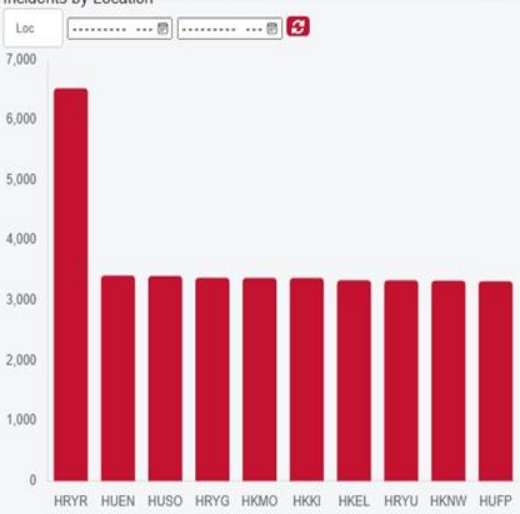
Geographic Distribution



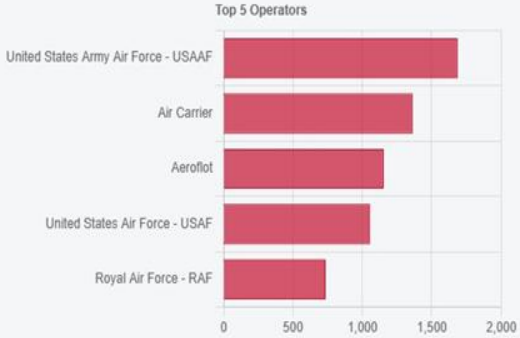
Incidents by Location



Recent Classification Results

Date	Operator	Aircraft	Phase	Classification	Confidence
2025-06-01	Unknown	Skyhawk 172/Cutlass 172	Climb	OTHER	80
2025-05-01	Personal	Skyhawk 172/Cutlass 172	Landing	RE	80
2025-05-01	Air Carrier	B737-800	Initial Climb	EQUIP	80
2025-05-01	Air Taxi	Citation III; VI; VII (C650)	Climb	OTHER	80
2025-05-01	Air Carrier	Commercial Fixed Wing	Unknown	GRND	80
2025-05-01	Air Carrier	A321	Parked	GRND	80
2025-05-01	Corporate	Caravan 208B	Cruise	OTHER	80
2025-05-01	Air Carrier	Regional Jet CL65; Undifferentiated or Other Model	Taxi	RINCUR	80
2025-05-01	Air Carrier	B787 Dreamliner Undifferentiated or Other Model	Cruise	EQUIP	80
2025-05-01	FBO	SR20	Initial Approach	OTHER	80

Incidents by Operator



Aircraft Type Distribution

SAFE AIR

Dashboard Evaluation

Evaluator: **RONNIE** Logout

SAFETY ANALYTICS

- Dashboard
- Reports
- Evaluation

2025-05-01	FBO	SR20	Initial Approach	OTHER	80

Incidents by Phase of Flight

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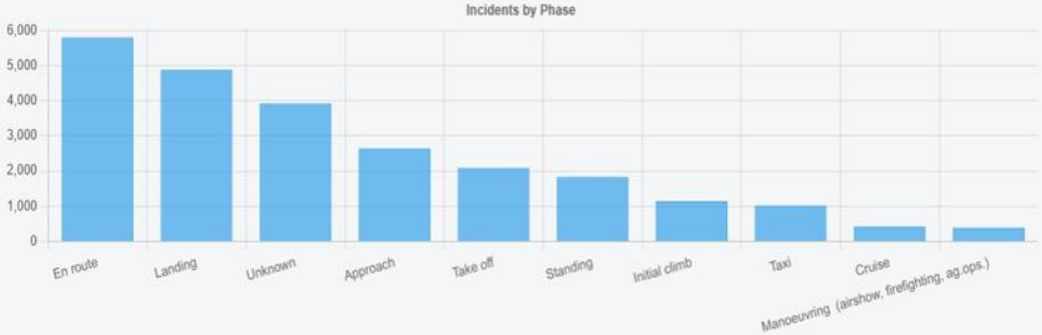
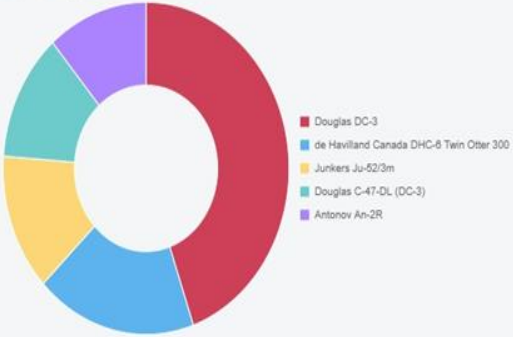


Aircraft Type Distribution



Seasonal Trends (Month vs Year)

Risk Heatmap (Phase vs Category)

Acronyms and Definitions

ICAO - International Civil Aviation Organisation
ADREP - Accident/Incident Reporting
ICAO Annex 19 -